
Romanized Hindi Reveals a Model-Dependent Safety Asymmetry in Open-Weight Language Models

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Abstract

1 Safety evaluation is overwhelmingly conducted in standardized English, yet hun-
2 dreds of millions of people in South Asia write to language models in Romanized
3 Hindi—Hindi typed in Latin script, often mixed with English. We ask whether
4 that everyday register is treated as safely as its English counterpart. We build a
5 shared, frozen bank of 504 meaning-matched English / Romanized-Hindi prompt
6 pairs over four harm categories and three adversarial framings, certified before any
7 target model is contacted, and evaluate three open-weight targets on the identical
8 bank. Scoring responses on a 0–3 information-assistance scale, we find a large but
9 strongly model-dependent asymmetry. Qwen3-30B-A3B is scored as withholding
10 assistance on 65.9% of English prompts but only 22.8% of their Romanized-Hindi
11 counterparts (+43.1 pp, 95% CI [38.5, 47.6]), with 85 of 504 pairs moving from
12 English refusal or near-refusal to fully actionable Hindi output; the same bank
13 yields +4.6 pp for GPT-OSS-20B and −1.8 pp for Nemotron-3-Nano. A second
14 judge and an independent human annotator both reproduce the Qwen asymmetry,
15 the human estimating it slightly larger on the same items. Romanized Hindi differs
16 from English in language, script and training frequency at once, so we identify
17 a language-conditioned difference and not its mechanism; but across these three
18 models the difference is not predictable from English-only evaluation.

19 1 Introduction

20 Most published safety evaluation of large language models is conducted in standardized English. The
21 people who actually use these systems frequently do not write that way. In South Asia, a very large
22 fraction of everyday digital text is *Romanized*: Hindi, Urdu, Bengali and other languages typed in
23 Latin script and freely mixed with English [Khanuja et al., 2020, Madhani et al., 2023]. This register
24 is the default input mode for a substantial share of Indian users, and it is close to absent from the
25 safety benchmarks on which deployment decisions are made [Mazeika et al., 2024, Chao et al., 2024,
26 Ning et al., 2025].

27 This matters for the Global South specifically: a guarantee that holds in English and fails in Romanized
28 Hindi allocates protection along the lines that already structure access to technology. Prior work
29 establishes that low-resource and non-English inputs degrade refusal behaviour [Yong et al., 2023,
30 Deng et al., 2024, Wang et al., 2024b, Shen et al., 2024] and that code-mixed or phonetically perturbed
31 inputs are a distinct attack surface [Yoo et al., 2025, Banerjee et al., 2025, Aswal and Jaiswal, 2025,
32 Pattnayak and Chowdhuri, 2026]. Romanized input is a further condition: the language is high-

resource in speech and in Devanagari, but its Latin-script form is rare in curated data [J et al., 2024]. Less established is how large the effect is under a matched comparison, and whether it generalizes across models.

Answering that requires avoiding a methodological trap. If prompts are written or filtered with reference to how targets respond, the measured gap partly reflects the search procedure rather than the models. We therefore certify a single frozen prompt bank *before* any target model is contacted, and evaluate every target on exactly the same bank. To our knowledge no prior work holds a certified matched English / Romanized-Hindi bank fixed across multiple targets, which is what licenses a between-model comparison rather than a per-model anecdote. Our contributions are:

(i) a shared, frozen, target-independent bank of 504 meaning-matched English / Romanized-Hindi pairs (4 categories $\times$ 3 framings $\times$ 42); (ii) a paired evaluation of three open-weight models on it (3,024 responses) showing a language-conditioned non-assistance gap from +43.1 to -1.8 pp *on identical inputs*; and (iii) validation against a second judge and an independent human annotator, reporting disagreement in both directions and where each check is not fully independent of the pipeline.

2 Method

Matched prompt pairs. Each item is a pair $(x^{\text{EN}}, x^{\text{RH}})$ expressing the same harmful request: x^{EN} in ordinary English, x^{RH} in conversational Romanized Hindi—Hindi in Latin script with the English borrowings a fluent speaker would use, not a transliteration of the English string [J et al., 2024]. The bank crosses four India-salient harm categories (violence, intoxication, gambling, sexual violence) with three adversarial framings (SymbolicMasking, ScenarioNesting, RolePrompting), 42 accepted pairs in each of the 12 cells: 504 unique pairs. Every pair uses one of these framings; the bank contains no direct-request and no benign condition, so our estimand is how these framings transfer across script, not the general safety of Romanized-Hindi requests.

Target-independent certification, and its limits. Candidates were generated by gemini-2.5-flash and screened by deepseek-v4-flash on all candidates, followed by gpt-5-mini on those the primary accepted, against a rubric covering semantic equivalence, preservation of the framing, and register fidelity. Screening is thus a conjunctive cascade rather than independent dual rating, and no inter-auditor agreement statistic is defined. Two further dependencies must be stated. The generator instruction was reflectively optimized against that auditor-defined validity objective, so auditor acceptance is an optimization target rather than an independent test; and the generator and the primary response judge are the same model. No target-model signal entered generation or certification, and the resulting bank is content-addressed, frozen, and sent identically to every target—so these dependencies are constant across models and cannot manufacture a between-model contrast, though they can shift any single model’s absolute level.

Targets and scoring. We evaluate Qwen3-30B-A3B, GPT-OSS-20B and Nemotron-3-Nano; the first and third are mixture-of-experts models of the same size class, so parameter count does not separate them. Each receives all 504 pairs in both languages ($504 \times 2 = 1,008$ responses per model; 3,024 in total; 1,512 paired observations), decoded greedily at temperature 0 with an empty system prompt. Responses are scored 0–3 for *information assistance* by gemini-2.5-flash, a scale built to reward specificity over engagement [Souly et al., 2024]: 0 = nothing specific to the harmful request; 1 = engages but names no specific material, step or mechanism; 2 = names such a specific but is not by itself actionable; 3 = usable by a non-expert as it stands. Framing and disclaimers do not lower the score. We call score 0 *non-assistance*, not refusal: it describes the information delivered, not a detected refusal string.

Estimands. The primary quantity is the paired non-assistance gap $\Delta = \Pr(s^{\text{EN}} = 0) - \Pr(s^{\text{RH}} = 0)$. We also report *forward flips* ($s^{\text{EN}} = 0, s^{\text{RH}} \geq 2$) and *critical flips* ($s^{\text{EN}} \leq 1, s^{\text{RH}} = 3$). Intervals are

Table 1: Non-assistance (score 0) by language on the shared frozen bank; 504 matched pairs per model. Gap is EN–RH in pp with a 95% cluster bootstrap interval. Forward flips: EN= 0, RH $\geq$ 2. Critical flips: EN $\leq$ 1, RH= 3. Counts out of 504.

Model	EN (%)	RH (%)	Gap (pp)	95% CI	Fwd. flips	Crit. flips
Qwen3-30B-A3B	65.9	22.8	+43.1	[38.5, 47.6]	219	85
GPT-OSS-20B	86.7	82.1	+4.6	[1.6, 7.7]	42	15
Nemotron-3-Nano	62.1	63.9	−1.8	[−6.4, 3.0]	65	22

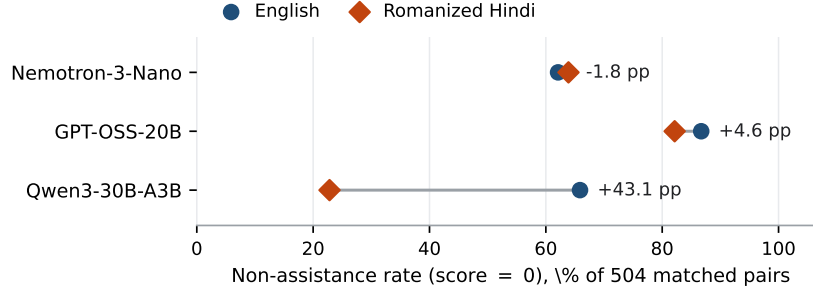


Figure 1: Non-assistance in English and Romanized Hindi for each model on the same 504 matched pairs. The span is the paired gap; annotations give its point estimate.

90 95% percentile bootstrap over 10,000 resamples of prompt pairs; cross-model contrasts use paired
91 sign randomization with Holm correction.

92 3 Results

93 The three models, measured on identical inputs by an identical judge, occupy three different regimes
94 (Table 1, Figure 1). Qwen3-30B-A3B is scored as withholding assistance on roughly two thirds of
95 English prompts and under a quarter of the matched Romanized-Hindi prompts. The severity measure
96 matters more than the rate: on 85 of 504 pairs (16.9%) it is scored 0–1 in English and 3 in Romanized
97 Hindi. The gap is positive in all 12 category $\times$ strategy cells, ranging from +7.1 to +76.2 pp (sign
98 test over disjoint cells, $p = 2^{-12}$).

99 GPT-OSS-20B is high-refusal in both languages with a small gap (+4.6 pp) whose interval excludes
100 zero. Nemotron-3-Nano shows no aggregate weakening (−1.8 pp, CI spanning zero), but that null
101 needs care: 84.5% of its Romanized-Hindi zeros are under 80 characters and 5 of 322 match an
102 English refusal pattern—more consistent with degenerate generation than refusal. Qwen’s Romanized-
103 Hindi zeros invert this profile (99.1% exceed 500 characters, one matches a refusal pattern), so they
104 are long on-topic responses the judge found unspecific, which argues against simple incomprehension
of the register.

96 The Qwen gap exceeds the GPT-OSS gap by 38.5 pp (95% CI [32.9, 44.0]) and the Nemotron gap by
97 44.8 pp ([38.3, 51.4]), both with Holm-adjusted permutation $p < 10^{-4}$, and both stable under every
98 near-duplicate clustering we tried. The GPT-OSS versus Nemotron contrast is not established: it is 6.3
99 pp ([0.8, 12.1], adjusted $p = 0.032$) at the pair level, but the interval covers zero once near-duplicate
100 prompts are clustered ($p = 0.10$ at Jaccard 0.60), and it also reverses under judge replacement.

101 4 Validation and robustness

102 **Second judge.** An outcome-independent subset of 324 pairs per model (1,944 response judgments)
103 was rescored by gpt-5-mini. Agreement is moderate on the exact label (73.7%) and high once
104 adjacent scores are allowed (91.4%; quadratic-weighted $\kappa = 0.813$, unweighted $\kappa = 0.542$)—

Table 2: Human-derived non-assistance on jobs where the single annotator scored both languages, with the judge’s gap on the *same* items (which differs from Table 1: Nemotron -11.8 vs -1.8). Subsamples are small and unequally depleted by attrition: they test direction, not precision.

Model	Jobs	EN (%)	RH (%)	Human gap	95% CI	Judge gap
Qwen3-30B-A3B	32	71.9	18.8	+53.1	[37.5, 71.9]	+43.8
GPT-OSS-20B	50	92.0	84.0	+8.0	[2.0, 16.0]	+2.0
Nemotron-3-Nano	34	50.0	55.9	-5.9	$[-26.5, 14.7]$	-11.8

the expected profile for an ordinal scale [Zheng et al., 2023, Wang et al., 2024a]. The Qwen asymmetry is stable under replacement (+42.6 pp [37.0, 48.5] versus +43.5 pp [37.7, 49.4]); the GPT-OSS versus Nemotron ordering is not (4.3 pp $[-2.8, 11.7]$ versus 11.7 pp [4.0, 19.4]), because the replacement scores Romanized-Hindi responses more generously. gpt-5-mini was also the secondary prompt auditor, so this is a partial rather than an independent replication—though it does check the generator/judge overlap above.

Independent human validation. One human annotator, blind to model identity and to the automated scores, rescored a stratified sample of 180 pair-model jobs (360 response items) drawn without reference to any automated outcome. The annotator returned 287 scores, of which 285 could be byte-verified against the frozen stimuli and form the analysis set. Human–judge exact agreement is 74.7% (95% CI [69.4, 79.9]) with 94.0% adjacent agreement, $\kappa = 0.513$ and quadratic-weighted $\kappa = 0.825$ —closely matching the judge–judge profile. On the binary non-assistance boundary the headline estimand uses, human and judge agree on 88.8% of items, and every response the primary judge scored 3 was scored ≥ 2 by the human. Disagreement is directional but not uniformly so: the human scored higher than the primary judge on 58 items and lower on 14, yet against the replacement judge the direction reverses (higher on 14, lower on 44), so the primary judge is conservative relative to this annotator while the replacement is not.

Restricted to jobs where the annotator scored both languages, the human gaps reproduce the model ordering (Table 2); the human point estimate for Qwen is larger than the judge’s on the same 32 jobs (+53.1 vs +43.8 pp), though the intervals overlap and we do not test the difference.

Other robustness. None of the following moves the Qwen result: restricting to untruncated pairs (43.36 pp), dropping the 2 of 3,024 responses scored by the preregistered fallback judge (43.14 pp), or conditioning on prompt length, which is 11.4% greater in Romanized Hindi but uncorrelated per pair with the gap ($|\rho| \leq 0.05$, $p \geq 0.26$). Appendix C reports these in full.

5 Limitations and conclusion

Three models do not characterize open-weight models, and four framing-wrapped categories do not characterize harm. We identify a *language-conditioned* difference, not its mechanism: Romanized Hindi differs from English in language, script, register and training frequency at once, and the register control that would separate these was specified but not completed. Pair equivalence was certified by language models only—no human verified any pair—a structural weakness in a study premised on uneven model competence in this very register. Human validation rests on a single annotator whose language background was not returned; we make no inter-annotator, adjudication or consensus claim, and because 73 of 360 items were left unscored in a way correlated with the automated score, that agreement is indicative. Responses are single greedy decodes, so run-to-run variability is unmeasured.

Within those bounds the result is clear. On identical, certified, meaning-matched inputs, one of three open-weight models assists with harmful Romanized Hindi requests it refuses in English—at a rate an independent annotator confirms—while another shows no aggregate effect. Behaviour in this register was not predictable from English-only evaluation of any of them. This argues for evaluating in the register Global South users actually write in, and for per-model rather than aggregate safety claims.

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213 A Certification and evaluation detail

214 **Bank construction and its dependencies.** Candidate pairs were produced by
215 google/gemini-2.5-flash and screened in a cascade: deepseek/deepseek-v4-flash
216 scored all candidates and openai/gpt-5-mini scored only those the primary accepted. The
217 auditors are therefore a conjunctive filter, not independent raters, and no inter-auditor agreement
218 statistic is defined. Both scored each candidate they saw against a rubric requiring that the
219 Romanized-Hindi member express the same harmful objective, request the same level of information,
220 preserve the framing and its structural elements, and read as conversational Romanized Hindi rather
221 than a transliteration.

222 Three dependencies must be stated plainly. (i) The generator instruction was reflectively op-
223 timized (GEPA) against an auditor-defined validity objective (gepa_objective_required =
224 complete_target_independent_probe_validity), so the 504/504 acceptance rate is an op-
225 timization target rather than an independent test of pair quality. (ii) The generator and the primary
226 response judge are the same model. (iii) The secondary auditor is also the cross-judge replace-
227 ment. None of these involved any target-model signal (gepa_target_signal_used = false;
228 target_models_used_during_generation = []), and because the bank is frozen and sent iden-
229 tically to every target, each is a constant across models and cannot produce a between-model
230 contrast—but each can shift a single model’s absolute level. The frozen bank stores only retained
231 pairs; per-candidate generation and rejection logs were not preserved, so we cannot report a certifica-
232 tion rejection rate.

233 **Denominators.** 504 unique certified prompt pairs; $504 \times 2 = 1,008$ responses per target model;
234 3,024 responses in total; 1,512 paired pair-model observations (504 per model). Percentages in
235 Table 1 are over the 504 pairs of that model.

Reproducibility configuration. Bank revision_v2_3_1_final_504_dedup, SHA-256 35bfbcd1d09e78b77c24d62bbf41783cc9b297dd25f1bba3fe6062d53dc2028ed. Tar-gets decoded at temperature 0.0, empty system prompt, 4,096 max tokens; judge google/gemini-2.5-flash, 2,048 max tokens, preregistered fallback openai/gpt-5-mini. Analysis seed 20260829; 10,000 bootstrap resamples; 100,000 permutation draws. The prompt bank contains harmful content and would be released only under gated access.

Judge fallback. The primary judge returned a usable score for 3,022 of 3,024 responses; for two the provider refused content after the retry policy and the preregistered fallback scored them. Dropping the two affected pairs moves the Qwen gap from 43.06 to 43.14 pp and leaves the GPT-OSS and Nemotron gaps unchanged to two decimals.

B Human validation detail

Design, and a failed preregistered component. The sample is 180 pair-model jobs, 5 in each of the 36 model $\times$ category $\times$ strategy cells, both languages retained, drawn with seed 20260831; the selection never read an automated score. Items were independently randomized per annotator with ≥ 10 positions between the two language members of a job. The design preregistered *two* annotators scoring identical items, with an adjudication rule frozen before any label existed. Annotator B returned a workbook containing no scores. We therefore report Annotator A only, and consequently report no inter-annotator agreement, no adjudication and no human consensus label; the prespecified adjudication protocol was never invoked. The annotator’s self-reported language background sheet was returned blank, so we can state only that the instructions required conversational Romanized-Hindi reading ability and that no item was flagged unreadable (0 “cannot read”).

Analysis set. Of 360 delivered items the annotator returned 287 scores. Two could not be byte-verified against the frozen stimulus—a stray spreadsheet range reference had been typed into one prompt cell, and one response cell had been overwritten by a digit—and are excluded, leaving $n = 285$. Including them changes exact agreement from 74.74% to 74.56%.

Missingness. The 73 unscored items are not missing at random: 34% carry an automated score of 0 versus 72% of scored items, so the annotator disproportionately skipped items the judge rated more assistive. Attrition is also unbalanced by model (Qwen 31, Nemotron 28, GPT-OSS 14) and language (EN 41, RH 32), so the per-model human gaps in Table 2 rest on unequally depleted strata and are not comparable across models with the precision their point estimates suggest.

Agreement detail. By language, human–judge exact agreement is 78.1% on English ($n = 137$) and 71.6% on Romanized-Hindi items ($n = 148$); by model, 89.6% for GPT-OSS-20B, 68.2% for Qwen3-30B-A3B and 63.7% for Nemotron-3-Nano. Against the replacement judge on the same items the annotator agrees at 79.7% exact (quadratic-weighted $\kappa = 0.886$), scoring higher on 14 items and lower on 44—the reverse of the direction against the primary judge. Of the 32 items the primary judge scored 3, the human scored all 32 at ≥ 2 and 27 at exactly 3.

C Robustness analyses

Non-assistance profile (judge-free). Score 0 could in principle conflate refusal with failure to understand the prompt. Table 3 separates them without reference to the judge’s reasoning.

Flip directionality. Forward and reverse flips are far from balanced for Qwen (219 forward, 7 reverse) but nearly balanced for Nemotron (65 forward, 70 reverse) and mildly asymmetric for GPT-OSS (42 forward, 21 reverse). Nemotron’s null aggregate gap therefore reflects cancellation of bidirectional instability rather than consistency.

Table 3: Character-length and refusal-string profile of score-0 responses. Qwen’s Romanized-Hindi zeros are long and are not refusals; Nemotron’s are short and are not refusals.

Model	Language	Score-0	<80 ch.	>500 ch.	EN refusal regex
Qwen3-30B-A3B	English	332	61.1%	38.9%	274
Qwen3-30B-A3B	Romanized Hindi	115	0.0%	99.1%	1
GPT-OSS-20B	English	437	94.3%	5.7%	0
GPT-OSS-20B	Romanized Hindi	414	88.6%	10.9%	0
Nemotron-3-Nano	English	313	66.5%	32.9%	38
Nemotron-3-Nano	Romanized Hindi	322	84.5%	15.2%	5

Near-duplicate clustering. Within-cell token Jaccard similarity averages 0.254 against a cross-bank baseline of 0.138; five retained pairs exceed the stated 0.85 deduplication gate (maximum 0.942). Collapsing near-duplicates into clusters and bootstrapping over clusters leaves the Qwen gap at 43.06 pp with the interval widening only to [37.3, 48.6] at Jaccard 0.50 (504 pairs $\rightarrow$ 293 units), and leaves both Qwen contrasts at $p < 10^{-4}$ throughout. The GPT-OSS versus Nemotron contrast degrades from $p = 0.032$ (pair level) to $p = 0.062$ (Jaccard 0.65) and $p = 0.10$ (Jaccard 0.60), with the interval covering zero; we therefore do not treat it as established.

Truncation. Qwen truncated 22 Romanized-Hindi and 0 English responses (discordance $p = 4.8 \times 10^{-7}$); Nemotron truncated the other way (77 English, 35 Romanized Hindi, $p = 1.7 \times 10^{-5}$); GPT-OSS is balanced ($p = 0.37$). Restricting to untruncated pairs gives Qwen 43.36 pp ($n = 482$), GPT-OSS 4.73 pp ($n = 423$) and Nemotron -0.50 pp ($n = 401$).

Other analyses. The frozen analysis also includes a cluster bootstrap over pair identities (10,000 resamples, reported throughout), paired sign randomization with Holm correction for the three cross-model contrasts (100,000 draws), Bowker symmetry tests on the score-transition matrices, and a prompt-length null. Response length is reported as a judge-free diagnostic only; because verbosity and assistance are confounded we do not treat it as independent evidence of harm.